

Project Title: NoteAI Lessons - Speech-to-Text Notes and AI Lesson Assistant

Selected Stack:

- Frontend: HTML
- Backend: PHP
- Database: Supabase
- NLP Tools: NLTK and spaCy
- Supporting Assets: CSS, Bootstrap, JavaScript

Project Idea:

The app works like cellphone notes. Students type or dictate lesson notes.

Every saved note is analyzed with NLP and stored in Supabase. The AI assistant then scans saved notes to answer direct lesson questions.

Implemented NLP Concepts:

1. Tokenization - NLTK displays tokenized note text.
2. Lemmatization - spaCy provides base forms for better search.
3. Sentiment Analysis - NLTK VADER classifies Positive, Negative, or Neutral.
4. Voice-to-Text - Dictate Note writes speech into the note editor.
5. Text-to-Speech - Speak Answer reads assistant output aloud.
6. Chatbot Assistant - scans saved notes and returns matching lessons.
7. Keyword Extraction - spaCy-assisted important term extraction.
8. Text Classification - Complaint, Inquiry, Feedback, Command, or General.

System Workflow:

Create note -> speech/text input -> PHP notes API -> Python NLTK/spaCy NLP -> Supabase notes table.

Ask question -> PHP chat API -> analyze query -> scan note keywords/lemmas/content -> return lesson answer.

Supabase Tables:

notes: id, title, content, tokens, lemmas, keywords, sentiment, sentiment_score, classification, entities, created_at, updated_at.

assistant_chats: id, question, answer, query_tokens, query_keywords, matched_note_ids, created_at.

How to Run:

1. Run database/nlp_assistant.sql in the Supabase SQL Editor.
2. Copy .env.example to .env and set SUPABASE_URL and SUPABASE_KEY.
3. python3 -m venv .venv && source .venv/bin/activate
4. pip install -r requirements.txt
5. python3 -m spacy download en_core_web_sm
6. php -S 127.0.0.1:8000
7. Open <http://127.0.0.1:8000>

Defense Demo Checklist:

Create a note by voice, show tokens/lemmas/sentiment/keywords, ask the assistant about the lesson, show matched notes, and